Find the Error

Grade 8 · 8.NS.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 0.333… (the 3 repeats forever) a rational number? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A repeating decimal can be written as a fraction. 0.333… = $\frac{1}{3}$, so it is rational.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $\sqrt{2}$ a rational number? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$\sqrt{2}$ is about 1.41421... and never repeats or ends. It is irrational.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 0.25 a rational number? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **0.25 = $\frac{1}{4}$, so it is rational. Every terminating decimal is rational.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Between which two whole numbers is $\sqrt{50}$? Student's answer: between 5 and 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$7^2 = 49$ and $8^2 = 64$. Since 50 is just above 49, $\sqrt{50}$ is between 7 and 8.**

Correct answer: **Between 7 and 8**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Locate $\sqrt{10}$ on a number line. Student says: It is closer to 4.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$\sqrt{10}$ is about 3.16. It is between 3 and 4 and is closer to 3.**

Correct answer: **Closer to 3**

Find the Error

Grade 8 · 8.NS.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\sqrt{7}$ and 2.6. Student says: $\sqrt{7}$ is less than 2.6.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2.6^2 = 6.76$, which is less than 7. So $\sqrt{7}$ is greater than 2.6.**

Correct answer: **$\sqrt{7}$ is greater**

Find the Error

Grade 8 · 8.EE.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $2^3 \times 2^2$. Student's answer: 2^6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied the exponents. For the same base, add the exponents: $2^3 \times 2^2 = 2^5 = 32$.**

Correct answer: **2^5 or 32**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 5^0. Student's answer: 0

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Any nonzero number to the power 0 equals 1. So $5^0 = 1$.**

Correct answer: **1**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Evaluate 10^{-2}. Student's answer: -100

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A negative exponent means the reciprocal. $10^{-2} = \left(\frac{1}{10}\right)^2 = \frac{1}{100} = 0.01$.**

Correct answer: **0.01**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find $\sqrt{49}$. Student's answer: 24.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided 49 by 2. The square root of 49 is 7, since 7 × 7 = 49.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 = 36$ for the positive value of x. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student took the square root of 9 instead of 36. Since $6^2 = 36$, $x = 6$.**

Correct answer: 6

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find $\sqrt[3]{27}$. Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student squared 3 instead of cubing it. 3 × 3 × 3 = 27, so the cube root is 3.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 4.5×10^3 in standard form. Student's answer: 450

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Multiplying by 10^3 moves the decimal three places right. $4.5 \times 1{,}000 = 4{,}500$.**

Correct answer: **4,500**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 0.00052 in scientific notation. Student writes: 5.2×10^4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The exponent should be negative because the number is small. The answer is 5.2×10^{-4}.**

Correct answer: **5.2×10^{-4}**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which is greater: 3×10^4 or 5×10^3? Student's answer: 5×10^3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$3 \times 10^4 = 30{,}000$, and $5 \times 10^3 = 5{,}000$. So 3×10^4 is greater.**

Correct answer: **3×10^4**

Find the Error

Grade 8 · 8.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A city has 2.4×10^5 people. How many people are in 3 cities of the same size? Student's answer: 7.2×10^6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student changed the exponent. $3 \times 2.4 = 7.2$, so the answer is 7.2×10^5.**

Correct answer: **7.2×10^5**

Find the Error

Grade 8 · 8.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide $(8 \times 10^6) \div (2 \times 10^2)$. Student's answer: 4×10^3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 2 from 6 incorrectly. 6 − 2 = 4, so the answer is 4×10^4.**

Correct answer: **4×10^4**

Find the Error

Grade 8 · 8.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(6 \times 10^{-3})(5 \times 10^{-2})$. Student's answer: 3×10^{-5}

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$6 \times 5 = 30$, and $-3 + (-2) = -5$. So $30 \times 10^{-5} = 3 \times 10^{-4}$.**

Correct answer: **3×10^{-4}**

Find the Error

Grade 8 · 8.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through $(0, 0)$ and $(2, 6)$. What is its slope? Student's answer: $\frac{1}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided run by rise. Slope = rise ÷ run = 6 ÷ 2 = 3.**

Correct answer: 3

Find the Error

Grade 8 · 8.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line rises 4 units for every 2 units across. What is its slope? Student's answer: $\frac{1}{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student flipped the ratio. Slope = 4 ÷ 2 = 2.**

Correct answer: 2

Find the Error

Grade 8 · 8.EE.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (1, 5) and (3, 1). What is its slope? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The line goes down as x increases, so the slope is negative. (1 − 5) ÷ (3 − 1) = −4 ÷ 2 = −2.**

Correct answer: **−2**

Find the Error

Grade 8 · 8.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, 0) and (1, 4). Write its equation. Student writes: y = x + 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is 4 and the y-intercept is 0. The equation is y = 4x.**

Correct answer: **y = 4x**

Find the Error

Grade 8 · 8.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line has a slope of 2 and a y-intercept of 3. Write its equation. Student writes: y = 3x + 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student swapped the slope and the intercept. m = 2 and b = 3, so y = 2x + 3.**

Correct answer: **y = 2x + 3**

Find the Error

Grade 8 · 8.EE.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line goes through (0, -2) and (2, 2). Write its equation. Student writes: y = -2x + 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is (2 - (-2)) ÷ (2 - 0) = 2, not -2. The equation is y = 2x - 2.**

Correct answer: **y = 2x - 2**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3(x + 2) = 2x + 9$. Student's answer: $x = 15$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$3x + 6 = 2x + 9$ gives $x = 3$. The student added the constants incorrectly.**

Correct answer: **$x = 3$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2x + 3 = 2x + 5$. Student's answer: $x = 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Subtracting 2x leaves 3 = 5, which is never true. The equation has no solution.**

Correct answer: **No solution**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $4(x + 1) = 4x + 4$. Student's answer: $x = 0$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both sides simplify to 4x + 4, which is always true. Every real number is a solution.**

Correct answer: **Infinitely many solutions**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x + 1$ and $y = 3x - 3$. Student's answer: $(1, 2)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x + 1 = 3x - 3$ gives $x = 2$, and then $y = 3$. The solution is $(2, 3)$. The student used $x = 1$.**

Correct answer: **$(2, 3)$**

Find the Error

Grade 8 · 8.EE.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $x + y = 10$ and $x - y = 4$. Student's answer: $(3, 7)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Adding gives $2x = 14$, so $x = 7$ and $y = 3$. The student swapped the values.**

Correct answer: **$(7, 3)$**

Find the Error

Grade 8 · 8.EE.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = 2x + 1$ and $y = 2x + 5$. Student's answer: (1, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Both lines have slope 2 but different intercepts, so they are parallel. There is no solution.**

Correct answer: **No solution**

Find the Error

Grade 8 · 8.F.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is the relation {(1, 2), (1, 3), (2, 4)} a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The input 1 is paired with both 2 and 3. A function gives exactly one output for each input.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is y = 2x a function? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each x gives exactly one y, so y = 2x is a function.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph crosses a vertical line in two places. Is the graph a function? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Two crossings mean one input has two outputs, so it fails the vertical line test. It is not a function.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A has a rate of change of 3 from a table. Function B is $y = 2x + 3$. Which has the greater rate of change? Student's answer: Function B

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Function A has a rate of 3, and Function B has a rate of 2. Function A is greater.**

Correct answer: **Function A**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A is y = 3x + 1. Function B has the points (0, 2) and (1, 5). Which has the greater starting value? Student's answer: Function A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Function A starts at 1. Function B starts at 2, so Function B has the greater starting value.**

Correct answer: **Function B**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Function A is y = 5x. Function B has the points (0, 0) and (1, 4). Which has the greater rate of change? Student's answer: Function B

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Function A has a rate of 5. Function B has a rate of 4. Function A is greater.**

Correct answer: **Function A**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find y when $y = x^2$ and $x = 3$. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied by 2. $3^2 = 9$, so y = 9.**

Correct answer: 9

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which function is nonlinear: $y = 4x + 1$ or $y = x^2$? Student's answer: $y = 4x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$y = 4x + 1$ has a constant rate of change, so it is linear. $y = x^2$ is nonlinear.**

Correct answer: **$y = x^2$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The table has x = 1, 2, 3 and y = 1, 4, 9. Is the relationship linear? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The changes are 3 and then 5. The rate is not constant, so it is not linear.**

Correct answer: No

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A table has the points (0, 3) and (2, 7). Write the linear function. Student writes: y = 3x + 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The rate of change is (7 − 3) ÷ 2 = 2, and the initial value is 3. The function is y = 2x + 3.**

Correct answer: **y = 2x + 3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the initial value of y = 5x - 4? Student's answer: 5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **5 is the rate of change. The initial value is -4, because y = -4 when x = 0.**

Correct answer: **-4**

Find the Error

Grade 8 · 8.F.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A tank has 10 gallons and loses 2 gallons each hour. Write the function. Student's answer: starting value 10 and rate 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The rate is the amount lost each hour, which is -2. The function is y = 10 - 2x.**

Correct answer: **Rate -2, starting value 10**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph starts high on the left and falls steadily to the right. Is it increasing or decreasing?

Student's answer: increasing

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The graph goes down as x increases, so it is decreasing.**

Correct answer: **Decreasing**

Find the Error

Grade 8 · 8.F.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A graph is flat for a while and then rises steadily. Which description fits the flat part? Student's answer: increasing at a constant rate

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A flat part shows zero rate of change, so the quantity is constant there.**

Correct answer: **Constant**

Find the Error

Grade 8 · 8.F.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A temperature graph rises from 40 to 80 and then falls to 50. When is the temperature decreasing? Student's answer: from 40 to 80

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The temperature rises from 40 to 80. It falls after reaching 80, when it decreases to 50.**

Correct answer: **After it reaches 80**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Translate the point (2, 3) right 4 units. Student's answer: (2, 7)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student moved up instead of right. Right 4 gives (6, 3).**

Correct answer: **(6, 3)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Reflect the point (2, 3) across the x-axis. Student's answer: (-2, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Reflecting across the x-axis changes the sign of y, not x. The answer is (2, -3).**

Correct answer: **(2, -3)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Reflect the point (2, 3) across the y-axis. Student's answer: (2, -3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Reflecting across the y-axis changes the sign of x. The answer is (-2, 3).**

Correct answer: **(-2, 3)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are two rectangles that are both 3 by 5 congruent, with one rotated? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Rotation keeps size and shape. The rectangles are congruent.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Are triangles with sides 3, 4, 5 and 4, 5, 6 congruent? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The side lengths do not match, so the triangles are not congruent.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle is reflected across a line to make a new triangle. Are they congruent? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A reflection keeps size and shape, so the triangles are congruent.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Dilate the point (3, -1) by a scale factor of 2 from the origin. Student's answer: (5, 1)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2 to each coordinate. Multiply: (6, -2).**

Correct answer: **(6, -2)**

Find the Error

Grade 8 · 8.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Dilate the point (6, 10) by a scale factor of $\frac{1}{2}$ from the origin. Student's answer: (3, 4)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student changed the y-value wrong. 10 × $\frac{1}{2}$ = 5, so the answer is (3, 5).**

Correct answer: **(3, 5)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A shape has a side of 2 units. It is dilated by a scale factor of 3. What is the new side length?

Student's answer: 5 units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 3 to 2. Multiply: 2 × 3 = 6 units.**

Correct answer: **6 units**

Find the Error

Grade 8 · 8.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Triangles have sides 3, 4, 5 and 6, 8, 10. Are they similar? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Each side in the second triangle is 2 times the first. The triangles are similar.**

Correct answer: **Yes**

Find the Error

Grade 8 · 8.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A side of 4 in one triangle matches a side of 6 in a similar triangle. The matching side is 10 in the first triangle. What is the matching side in the second? Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The ratio is $\frac{4}{6} = \frac{2}{3}$. So $10/x = \frac{2}{3}$, and $x = 15$.**

Correct answer: **15**

Find the Error

Grade 8 · 8.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rectangles are 2 by 3 and 4 by 6. Are they similar? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$. **The rectangles are similar.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has angles 50° and 60°. What is the third angle? Student's answer: 80°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted incorrectly. 180 − 110 = 70°.**

Correct answer: **70°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two parallel lines are cut by a transversal. One alternate interior angle is 65°. What is the other?

Student's answer: 115°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Alternate interior angles are equal. The other angle is 65°.**

Correct answer: 65°

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Parallel lines are cut by a transversal. A corresponding angle is 40°. What is the matching corresponding angle? Student's answer: 140°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Corresponding angles are equal. The matching angle is 40°.**

Correct answer: **40°**

Find the Error

Grade 8 · 8.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. $3^2 + 4^2 = 25$, so c = 5.**

Correct answer: 5

Find the Error

Grade 8 · 8.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has legs 6 and 8. What is the hypotenuse? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the legs. $6^2 + 8^2 = 100$, so c = 10.**

Correct answer: **10**

Find the Error

Grade 8 · 8.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A right triangle has a hypotenuse of 13 and a leg of 5. What is the other leg? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 5 from 13. $13^2 - 5^2 = 169 - 25 = 144$, so the leg is 12.**

Correct answer: **12**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangular floor is 12 feet by 5 feet. What is the length of the diagonal? Student's answer: 17 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. $12^2 + 5^2 = 169$, so the diagonal is 13 feet.**

Correct answer: **13 feet**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A ladder 10 feet long leans on a wall with its base 6 feet from the wall. How high does it reach?

Student's answer: 16 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 10 and 6. $10^2 - 6^2 = 64$, so the height is 8 feet.**

Correct answer: **8 feet**

Find the Error

Grade 8 · 8.G.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box is 3 by 4 by 12 units. What is the length of its space diagonal? Student's answer: 19

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. The space diagonal is $\sqrt{(3^2 + 4^2 + 12^2)} = \sqrt{169} = 13$ units.**

Correct answer: **13 units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (0, 0) and (6, 8). Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 6 and 8. The distance is $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$.**

Correct answer: **10**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (1, 1) and (4, 5). Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 3 and 4. The distance is $\sqrt{(3^2 + 4^2)} = 5$.**

Correct answer: 5

Find the Error

Grade 8 · 8.G.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the distance between (-2, 3) and (1, -1). Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The horizontal distance is 3 and the vertical distance is 4. The distance is $\sqrt{(9 + 16)} = 5$.**

Correct answer: 5

Find the Error

Grade 8 · 8.G.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cylinder with radius 2 and height 3. Student's answer: 6π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot to square the radius. $V = \pi \times 2^2 \times 3 = 12\pi$, which is about 37.7 cubic units.**

Correct answer: **12π, about 37.7**

Find the Error

Grade 8 · 8.G.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a cone with radius 3 and height 4. Student's answer: 36π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot the $\frac{1}{3}$. $V = (\frac{1}{3}) \times \pi \times 9 \times 4 = 12\pi$, which is about 37.7 cubic units.**

Correct answer: **12π, about 37.7**

Find the Error

Grade 8 · 8.G.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the volume of a sphere with radius 3. Student's answer: 12π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not cube the radius. $V = \left(\frac{4}{3}\right) \times \pi \times 27 = 36\pi$, which is about 113.1 cubic units.**

Correct answer: **36π, about 113.1**